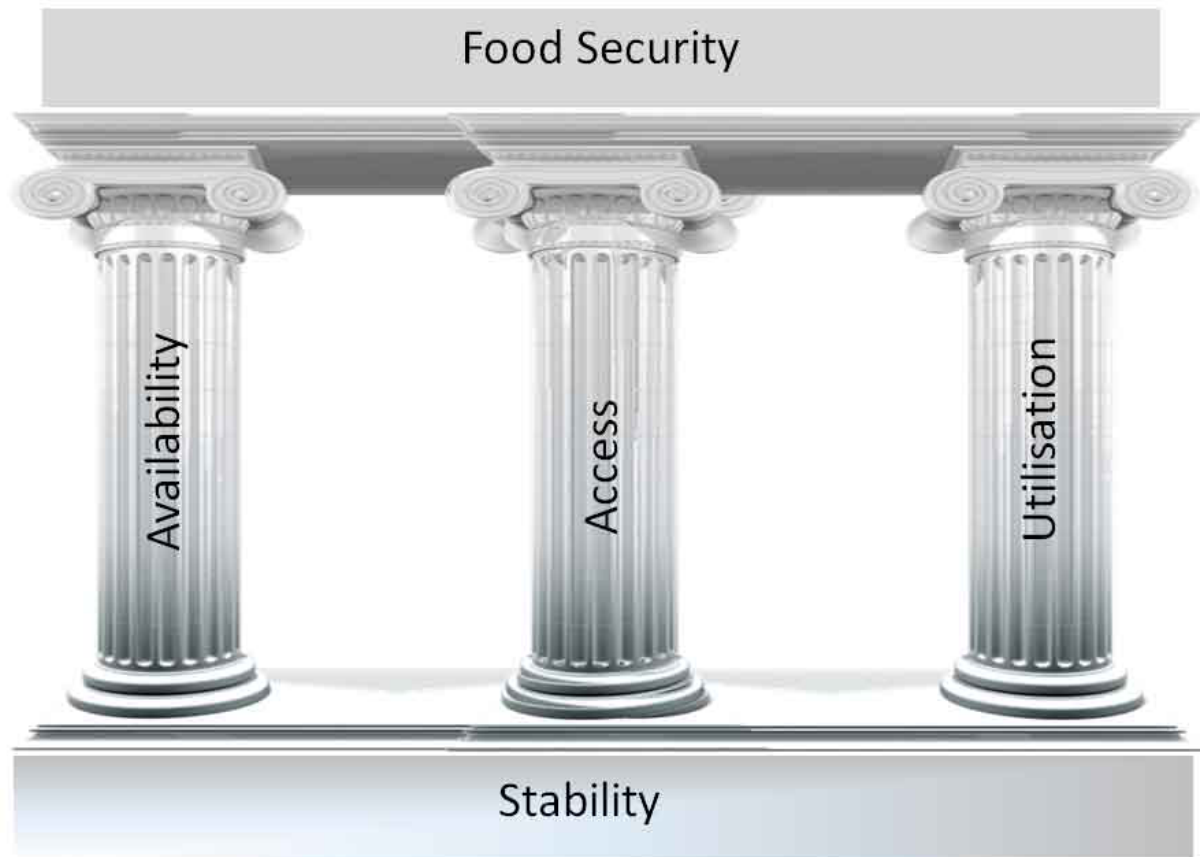


INTRODUCTION TO HUMAN NUTRITION AND DIETETICS (NUD 201)



Prof. O. S. IJAROTIMI

Mrs Kareem

Basic nutrition principles with special emphasis on nutrients (protein, carbohydrate, fat, minerals, vitamins, and water); sources, functions, digestion and absorption and their problems (signs and symptom of deficiency diseases).

Expected learning outcomes

1. discuss basic nutrition principles;
2. describe the concept of digestion, absorption and functions of nutrients;
3. discuss principles of energy balance;

Mrs Adeyemo

An in-depth discussion on foods, their characteristics, effects of anti-nutrients, and nutrient – nutrient and nutrient-drug interactions.

Expected learning outcomes

4. explain the determinants of food choices;
5. explain the effects of anti-nutrients and nutrient-nutrient interaction;
6. discuss with dietary guidelines;
8. explain the concept of dietary diversity;

Prof. Ijarotimi

Recommended dietary and nutrient intakes, safe levels of nutrient intake, balance studies to determine nutrient requirements will be addressed.

Expected learning outcomes

7. give an overview of food and nutrition security;
9. identify the non-conventional foods;
10. discuss the concept of food enrichment, supplementation and fortification; and
11. give an overview of basic nutrition evaluation methods.

Food security and Nutrition security

The term food and nutrition security complement the terms food security and food safety, which are defined as follows:

Term	Definition
Food and Nutrition Security	Physical, social, and economic access at all times to sufficient, safe, and nutritious food that meets one's nutritional needs and an environment with adequate sanitation and access to health and care services.
Food Security	Physical, social, and economic access at all times to sufficient, safe, and nutritious food that meets one's nutritional needs.
Food Safety	Unsafe foods may contain bacteria, parasites or chemical substances that are harmful. They cause diseases such as diarrhea or cancer. Food safety must therefore be ensured during the production and processing of food in order to prevent food contamination.

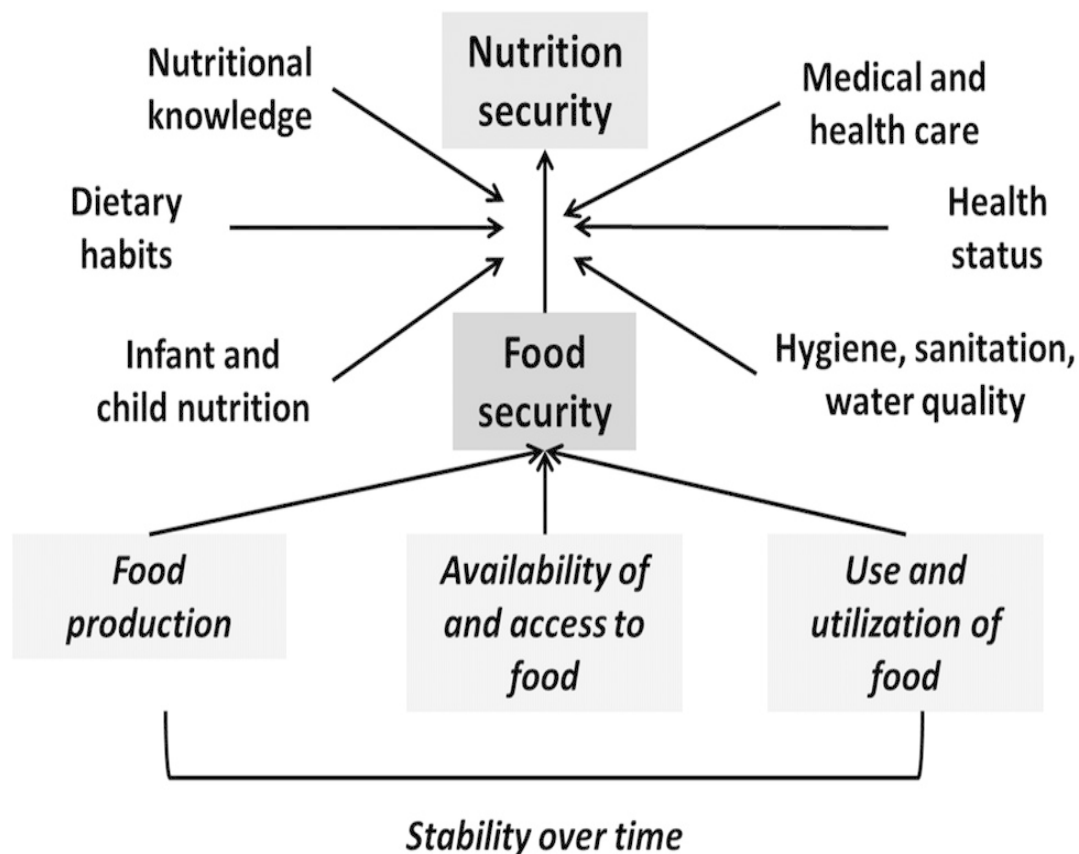


Fig. Conceptual relationship between food and nutrition security

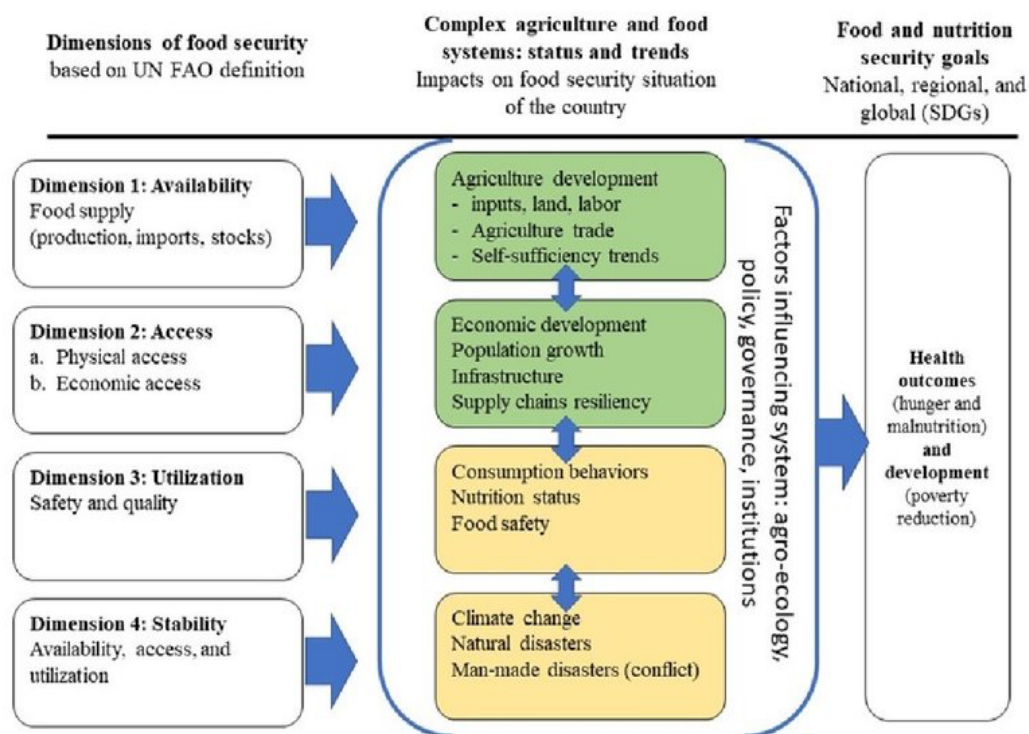
Four Dimensions of Food Security

Availability: It is mainly the supply or production of food materials. “The amount of food that is present in a country or area through all forms of domestic production, imports, food stocks and food aid”.

Accessibility: The concept of accessibility in terms of food security refers to “physical, social and economic access”. This term was first introduced by the Amartya Sen in the early 1980s after the Bengal famines. Accessibility means affordability in which every household member, at all times, have access to enough food for an active, healthy life.

Utilization: It means to the proper consumption of food and nutrition. No access to clean drinking water, poor environment, lack of hygiene, poor health infrastructure, lead to reduced assimilation of the consumed food. To achieve nutritional security, environmental hygiene necessary, Primary health care and clean drinking water facility should be adequately available. Cooking habits also need cautious evaluation because some methods of cooking may lead to the loss of essential nutrients Swaminathan.

Stability: Stability defines the temporal dimension of food and nutrition security, individually the time frame in which food security is being considered. Stability is assumed when the supply food, income and economic resources remain constant in a household level during the year and in the long- term. Besides, it is also essential to minimize external risks such as natural disaster, climate change, price volatility, conflicts or epidemics, etc.



Causes of Food Insecurity

There are many causes of food insecurity in Africa. This presentation will limit the scope to that of Nigeria as presented in the above where it has been discovered that the food insecurity in on the rise since 2009 in the country which will give a template to the other African countries due to the fact that countries in Africa face similar challenges but only differ in magnitude. Some of the popular and common causes are itemized below:

Causes of Food Insecurity

The causes of food insecurity

1. Poverty

Poverty is the number one driver of food insecurity, hindering subsistence farmers with small farms from investing in their farming practices so they can not only produce enough to eat, but also enough to sell their surplus at a profit. As a result, these farmers are forced to use less efficient agricultural methods without the benefit of climate adaptive tools or supplies, which limits the production of food and perpetuates the problem of poverty. When these farmers overuse their already depleted soil, attempting to grow enough to eat, they unknowingly exacerbate the problem. This is why World Renew prioritizes coming alongside farmers with the education and tools they need to farm sustainably and eventually reach financial stability.

2. Natural disasters

Floods, drought, typhoons, amongst other natural disasters have been known to wipe out whole crops, or even completely destroy farms – which is particularly devastating for rural communities and families who depend on these harvests for food.

3. War, Conflict and Political Instability

Of recent, the greatest threat to food security in Nigeria comes from the insurgency. As it is generally known that like other countries of the world, Nigeria is passing through one of her greatest challenges since independence from Britain in 1960. Of the six geo-political zones in Nigeria, the North East which is one of the critical zones that do not only provide the staple foods like grains, wheat and others but that is equally responsible for the provision of greater percentage of dairy products and animal protein in form of meat is currently under siege by insurgents. This has disrupted the agricultural activities in the areas while businesses worth millions of US Dollars have equally been halted. With this, the food security of the area is not only affected but almost every part of the country thus resulting to upward rising in the prices of food commodities in Nigeria. More than one million people have been displaced both internally and externally with more than ten thousand deaths. The remaining people in the affected areas are unable to continue their farming activities in the areas which had resulted in the alteration in the agricultural value chain in the country thereby resulting in reduction of food production. The problems started in 2009 and got escalated in the year 2014.

4. Urbanization

Like other countries of the world, increasing rural-urban migration due to urbanization play key roles in the emerging food insecurity in Nigeria. According to Food and Agricultural Organization (FAO), by year 2050, 70 percent of the world population is expected to be living in cities. By this, the agricultural production will be disrupted thereby increasing the food insecurity.

5. Population Growth

This is another factor responsible for food insecurity in Nigeria. Currently, Nigeria is the 6th populous country in the world with the estimated population of about 178 million people with annual growth rate of 3 percent. This has increased the demand for food products just like other countries of the world. With this glaring increase in the population, there is no commensurable increase in the agricultural production. According to the United Nations, Nigeria is expected to become third most populous country in the world by 2050 overtaking America.

6. Poor Agricultural Sector Development

This is another problem that poses great threat to food security in Nigeria and Africa as whole. Most African countries including Nigeria do not have sustainable Agricultural policy that will enhance food security in the long run. This is so because in Nigeria and majority of African countries as a whole until recently do not have stable leadership thereby resulting to policy somersault. Nigeria for example has not been on stable political leadership until 1999 when the country returned to full democracy devoid of military intervention. This has led to inconsistencies in policies that could have placed the country on a stable track that could have ensured that there is uninterrupted food supply. However, the opposite is the case as successive governments have engaged in different agricultural policies that have led to cancellation of the existing ones even when the extant ones proved prosperous and sustainable. Some government policies interfere with markets, create standards that inhibit trade and remove price signals to farmers. Export restrictions and trading bans isolate local markets and give farmers little incentive to expand production for the next season, limiting the potential supply response to soaring prices.

7. Climate Change

Climate Change is another reason why there is food shortage in Africa. It has changed the productivity pattern. The rain and water are less predictable now than before. The rain comes either too late or too early or for a shorter period. Farmers are confused and do not know when to cultivate their grains and other vegetables. Some years, the rain comes too early and when they plant the grains, the rain stops and the grains rot under the ground.

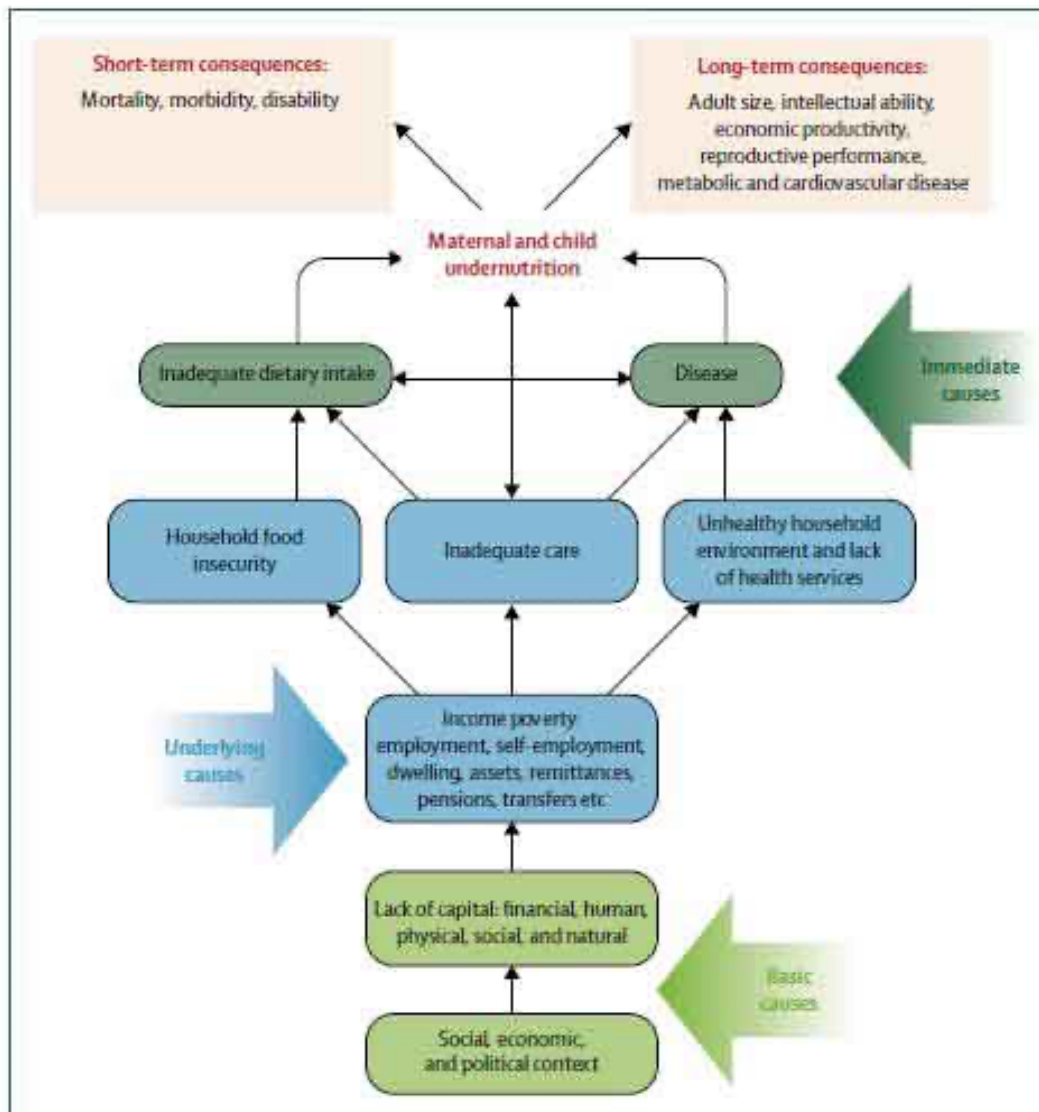


Fig. Consequences of Food Insecurity

Solutions to the Problems of Food Insecurity

In order to mitigate the effects of food insecurity on the people there should be proactive actions on the side of the leaders with a view to protecting their people from starving to death as we are currently witnessing in some African countries including Nigeria. To achieve this among other things, the following actions should be taken:

Solutions to Improve Food Security

1. Fight climate change

Climate change has a serious impact on our lives and food security. Implementing strategies like crop rotation and drip irrigation can increase yields, raise profit margins and lower production costs. Having these adaptive features in place also allows farmers to reduce weather losses and mitigating quality, which reduces greenhouse gas emissions in the long-run.

World Renew and the Canadian Foodgrains Bank work together with farmers to discover creative solutions to the problem by using simple yet innovative irrigation models and water retention. Combined with the principles of agricultural conservation – a climate-resilient farming method that will give farmers a stable and healthy food supply even in dry seasons.

2. Promote Crop diversification

Focusing on a single food crop or staple can lead to horrible results, such as blights and other plant disease outbreaks. In rural and remote communities that depend on locally produced crops, these outbreaks can have a detrimental impact on their food security. Diversifying crops will allow these families to mitigate the risks of plant disease outbreaks while also helping them achieve a more balanced diet. You can learn more about how crop diversification practices helped those like Maria Macz [here](#).

3. Improve infrastructure in rural community

Some farmers struggle to sell their products at the market due to improper infrastructure, such as broken roads, lack of storage, and inadequate food processing equipment. These issues cause farmers' crops to rot instead of being distributed to nearby communities. By improving the infrastructure, we can help distribute more food throughout different areas and lower food insecurity and food waste.

4. Consistence in Government Agricultural Policy Framework

First and foremost, the leaders must adhere strictly to unique agricultural policy that will be subjected to periodic review by experts concerned instead of the current policy somersault as we are currently witnessing. Once this is achieved, the successive governments will be able to pursue the food security with the desired vigour.

5. Provision of Infrastructures

Provision of social amenities such as roads, portable water, electricity, etc. should be provided to the rural dwellers to prevent or reduce rural-urban migration as this will ensure there is adequate labour for agricultural activities in the rural areas thereby increasing food productivities that will not only guarantee food security but will equally provide employment opportunities to the growing population in a way that it will reduce the pressure in the urban centers. Under this, the government should also encourage mechanized farming as a way of reducing the incidence of food insecurity as all over the world now agriculture is a serious business enterprise and the world has moved beyond the era of subsistence farming whereby agriculture is seen as feeding the immediate family with little or nothing for income drive. Modern agriculture is not only feeding the family but the world and is generating millions of employments and veritable source of income to both the farmers and source of foreign exchange to many countries of the world.

6. Population Control

Just like other African countries where poverty is rampant, population is majorly uncontrolled as the methods of achieving this menace such as contraceptives is still unpopular especially in the rural areas thereby resulting to population surge among the people thus encouraging undue competition for insufficient food. Government and other development partners such as world bank,

UNICEF, FAO, WFP and others should rise up to assist the government in educating people on birth control so that there can be food security guaranty.

7. Provision of Storage Facilities

To avoid waste that usually accompany harvest season in Nigeria, government should do everything possible to ensure there is adequate provision of processing and storage facilities for Agricultural products so that there will be all year-round food security and in turn boost the income level of farmers thereby reducing poverty which is the ultimate goal of the Nigerian government just like any other African government. The current situation in Nigerian has become so worrisome to the extent that vast majority of the harvested crops waste away during the farm season while huge amount of money is being used by the government to offset the import bills of the same commodities during the off season.

8. Abolition of Trade Barriers

There should be free trade across the African borders to guarantee free flow of food commodities across borders. Though, there are various extant trade agreements with other African countries especially the western Africa but the enforcement across nations has been very weak. If these trade agreements are fully operational the food insecurity will not only be minimized in Nigeria but in other African countries.

Impacts of Food Insecurity

1. Moral and Humanitarian Threat: Hunger, poverty and disease are interlinked and are direct result of lack of sustainable food supply / availability. Hunger reduces the natural defences against most diseases and is the main risk factor for illness worldwide. People living in poverty often cannot produce or buy enough food to eat and so are more susceptible to disease. Sick people are less able to work or produce food. The UN Standing Committee on Nutrition concluded that nutrition is an essential foundation for poverty alleviation and also for meeting MDGs related to improved education, gender equality, child mortality, maternal health and disease. Hunger is a major constraint to a country's immediate and long term economic, social and political development. The search for sustainable food has thrown man into acute deforestation which has resulted to serious environmental, climatic and health problems.

2. Developmental Threat: This is erasing the economic gains of the past decades, while putting at risk the recent historic investments in public health and nutrition, improved education and community development in poor countries. Without effective action to reverse these trends, developing countries could see a disabled generation, stunted both physically and mentally and chronically in need of assistance.

3. Strategic Threat: This is endangering the stability of developing countries due to rising food prices and fuel prices. The surge in prices has reduced the purchasing power of poor people and inhibited the ability of poor countries to import food for their hard-pressed population.

4. High prevalence of malnutrition

In developing countries, persistent malnutrition leaves children weak, vulnerable, and less able to fight such common childhood illnesses as diarrhea, acute respiratory infections, malaria, and measles. Adolescents and adults also suffer adverse consequences of food insecurity and malnutrition. Malnutrition can lead to decreased energy levels, delayed maturation, growth failure, impaired cognitive ability, diminished capacity to learn, decreased ability to resist infections and illnesses, shortened life expectancy, increased maternal mortality, and low birth weight.

5. Political instability

Food insecurity may also result to political instability just as food-insecure individuals may manifest feelings of alienation, powerlessness, stress, and anxiety, and they may experience reduced productivity, reduced work and school performance, and reduced income earnings. Household dynamics may become disrupted because of a preoccupation with obtaining food, which may lead to anger, pessimism, and irritation among other vices.

Effects of Improved Food Security

These common food insecurity solutions could contribute to feeding millions of people worldwide. Additionally, reducing food distribution middlemen would transfer more money to farmers' hands. This would give farmers more resources in their own community to provide relief from malnutrition with more diverse diets, which will mitigate high mortality rates.

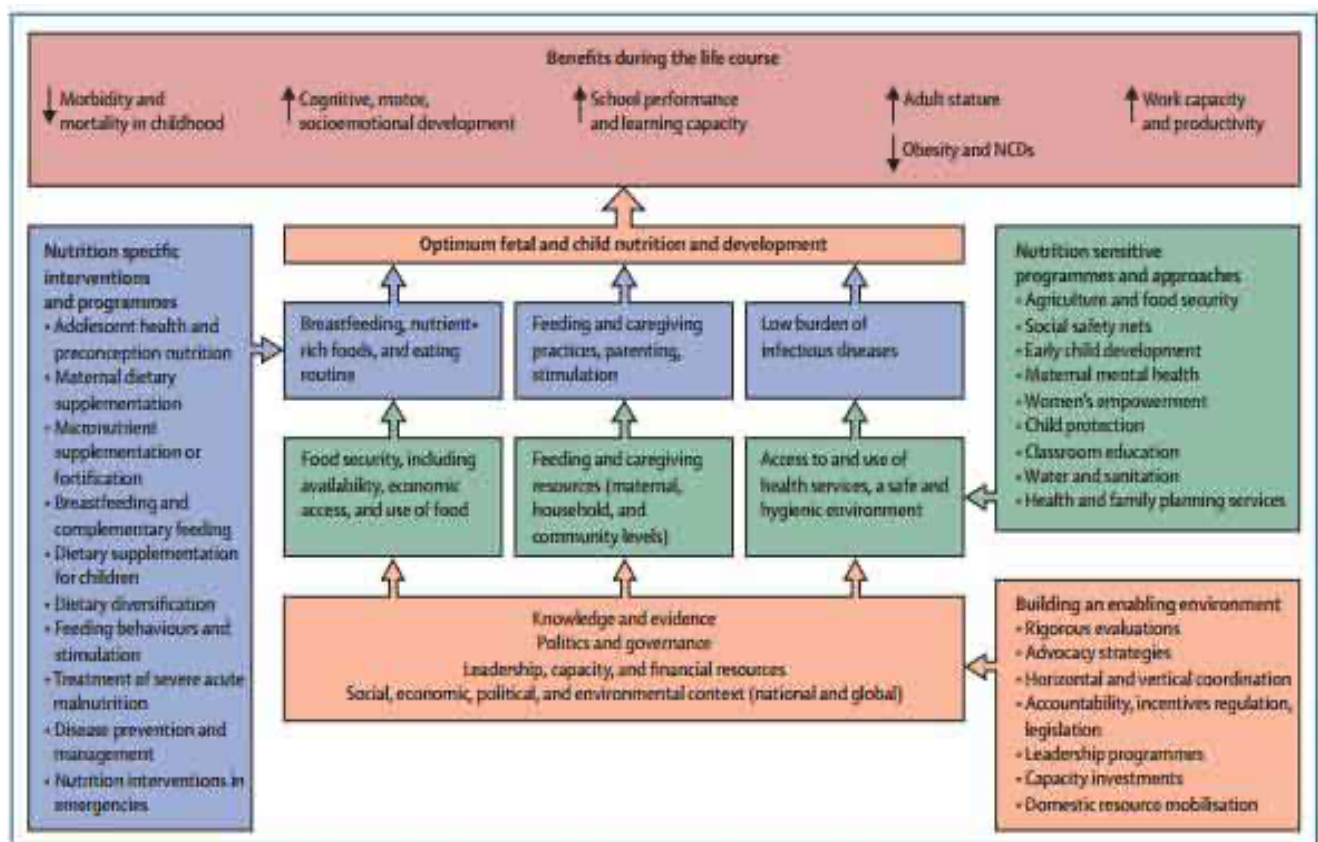


Fig. Benefits of food security

Conventional and Non-conventional Foods

Conventional foods are foods “consumed primarily for taste, aroma, and nutritive value.” Whether a product is intended for use as a conventional food depends, among other things, on the product’s ingredients and how the product is represented for use (which may be inferred by a variety of factors, including the product name, the product label, and even the product’s purported health benefits).

Non-conventional vegetables are of great importance for human feeding since they may provide vitamins, dietary fiber, carbohydrates, minerals and proteins. These foods are considered non-conventional because they are currently consumed by only a few people, usually in restricted areas or communities. In addition, non-conventional vegetables have sensory characteristics that makes their consumption easier), and they can be used in raw salads, soups, puree and omelets. The intake of non-conventional vegetables is associated with antioxidant, anti-inflammatory, antimicrobial and anti-cancer effects, that are partially attributed to the biological activities of their phytochemical constituents such as phenolic compounds, vitamins, carotenoids, flavonoids and minerals.

Non- conventional food may be microbes (Bacteria, Fungi), algae, insects etc. It may be used as whole body (parts). Scientists discovered microbes are rich in proteins, carbohydrates, vitamins, fats, minerals and essential amino acids and also used as food supplements. The concept of single cell protein (SCP) is central to non - conventional food. SCP refers to edible unicellular protein rich microorganisms. SCP may be fungi, yeast, algae, and bacteria. For examples spirulina is a cyanobacteria (blue- green algae) and well known for high protein content. Other example is Chlorella which is green algae and also content proteins, minerals, vitamins etc. Both (Spirulina & Chlorella) are used as food supplement currently. Now India is grown-ups in the field of large-scale production of Spirulina and Chlorella. Not only this, in many countries insects used as part of food. Some of the insects are beneficial for dietary source of proteins, carbohydrates, vitamins, fats, minerals and essential amino acids. Its larva and adult stages are more beneficial. In the case of bacteria, research is going on.

Types of Nonconventional Foods

Nonconventional foods (NCF) are some foods of plants or animals that people do not eat but which have been found by scientist to be edible and nutritious. These are of two types.

1. **Natural nonconventional foods-** These foods are obtained from nature. Some of the examples of nonconventional foods-

Khejri beans (*Prosopis cineraria*)- They are rich in proteins (15-18%), contains many kinds of minerals like phosphorus, iron, magnesium etc. they are one of the rich source of phosphorus. Pinju- They are not only rich source of proteins but also rich in phosphorus. They contain vitamins C, beta-carotene.

Karchi- They contains high amount of fat (13%) and also rich in protein and calcium.

B. Karchi (*Tribulus terrestris*)- They are rich source of phosphorus and iron.

Gullar (*Ficus glomerata*)- They are rich in calcium and zinc. They contains about 15 times more calcium than wheat.

2. Synthetic nonconventional foods- These are produced from chemicals like carbohydrates, hydrocarbons, organic as well as various inorganic matters by various processes like food technology, microbiology, genetic engineering or from existing natural products containing carbohydrates, proteins, fats, by various physical, chemical, microbiological or enzymatic modifications. Some of the examples of synthetic nonconventional foods-

1. Single cell protein- Single cell protein refers to sources of mixed protein extracted from pure or mixed cultures of algae, yeast, fungi or bacteria (grown on agricultural waste) used as a substitute for protein rich foods, in human and animal feeds. Since 2500 BC yeast has been used in bread and beverages or their products. In 1781 processes for preparing highly concentrated forms of yeast were established. Single cell proteins develop when microbes ferment waste materials (including woods, straw and food processing wastes, residues from alcohol production, hydrocarbons or human or animal excreta). But extraction was costly and usually less than 5%. Engineers were emphasis on increasing the concentration by various processes like centrifugation, precipitation etc. the various microbes used are *saccharomyces arevisae*, *aspergillus orygae*, *trichoderma*, *rhodopseudomonas capsulate*, *spirulina* etc.

2. Food pills- These are synthetic foods. These look like weak corn syrup and taste is also like corn syrup. Now a day these are widely used in space food. They contain various kinds of amino acids (l-Lysine HCL, l-Leucine, l-Isolucine, l-Phenylalanine, l-Arginine HCL, l-Mtheonine, l-Glutamine); water soluble vitamins (Thiamine, riboflavin, pyridoxine, biotin, folic acid, ascorbic acid, choline bitartrate); fat soluble vitamins (vitamin D, vitamin A, alfa-tocopherol acetate); salts (potassium iodide, magnesium acetate, cupric acetate, potassium hydroxide, magnesium oxide, sodium chloride, sodium benzoate) and carbohydrates.

3. Leaf protein concentrate- Leaf protein concentrate is a concentrated form of the proteins found in the leaves of plants. It has been examined as a human or animal food source, because it is potentially the cheapest, most abundant source of available protein. Although human derive some proteins directly from green leafy vegetables but the fibers are not digested by the human digestive system. Leaf protein concentrate is a good source of amino acids, polyphenyls, high fiber content but has anti nutritional factors like cyanide etc.

4. Seed meal-

- Cotton seed meal- Cotton seed meal is the ground meal from cotton seeds after the cotton seed oil has been extracted.
- Flax seed meal- They are highly nutritious and used in recipes. They have a pleasantly nutty taste. But they are needed to be ground to obtain the whole nutritional items.
- Tea seed meal- they are called as tea seed cake and tea seed powder which is a kind of residue of oil extract. They look like small rock about 10-15cm in diameter.

These nonconventional foods have great potential to fulfill the nutritional needs of human beings. From recent years food scientists were giving emphasis on the production of various kinds of products from these nonconventional foods and now many products can be found in the markets like seed meals, food pills etc. as population increases day by day and conventional foods are not sufficient to fulfill the hunger of people so we must have to give importance to these NATURE'S

SPECIAL FOODS. Apart from this, public are more health conscious so undoubtedly this food will play an important role in future.

Functional foods and Nutraceuticals

Each category plays a distinct role in dietary practices, yet they are interconnected by their contribution to health enhancement and disease prevention.

Functional foods: These are foods or food components that provide a health benefit beyond basic nutrition. For example, fortified dairy products and citrus fruits enriched with added nutrients fall into this category. Functional foods are part of the daily diet and are designed to improve health or prevent diseases. They include ingredients like polyunsaturated fatty acids and phytochemicals, which interact with the gut microbiota to influence health positively. These foods are not just nourishing; they also play specific roles in enhancing health through their bioactive components.

Nutraceuticals: The term “nutraceutical,” blends “nutrition” and “pharmaceutical,” and refers to foods or food parts that offer medical or health benefits, including disease prevention or treatment. Unlike general functional foods, nutraceuticals are used specifically for their purported medicinal effects. They might be consumed as regular foods or be part of a meal and are characterized by their ability to prevent or treat diseases. Examples include vitamins E and D, selenium, green tea, soy, and lycopene, which have been studied for their therapeutic potential in contexts like cancer therapy.

Examples of functional foods

Functional foods are generally separated into two categories: conventional and modified.

Conventional foods are natural, whole-food ingredients that are rich in important nutrients like vitamins, minerals, antioxidants, and heart-healthy fats. Meanwhile, modified foods have been fortified with additional ingredients, such as vitamins, minerals, probiotics, or fiber, to increase a food's health benefits.

Here are some examples of conventional functional foods:

- **Fruits:** berries, kiwi, pears, peaches, apples, oranges, bananas
- **Vegetables:** broccoli, cauliflower, kale, spinach, zucchini
- **Nuts:** almonds, cashews, pistachios, macadamia nuts, Brazil nuts
- **Seeds:** chia seeds, flax seeds, hemp seeds, pumpkin seeds
- **Legumes:** black beans, chickpeas, navy beans, lentils
- **Whole grains:** oats, barley, buckwheat, brown rice, couscous
- **Seafood:** salmon, sardines, anchovies, mackerel, cod
- **Fermented foods:** tempeh, kombucha, kimchi, kefir, sauerkraut
- **Herbs and spices:** turmeric, cinnamon, ginger, cayenne pepper
- **Beverages:** coffee, green tea, black tea

Here are some examples of modified functional foods:

- fortified juices
- fortified dairy products, such as milk and yogurt
- fortified milk alternatives, such as almond, rice, coconut, and cashew milk
- fortified grains, such as bread and pasta

- fortified cereal and granola
- fortified eggs

Potential Benefits of Functional Foods

Functional foods are associated with several potential health benefits.

Prevent nutrient deficiencies

Functional foods are typically high in important nutrients, including vitamins, minerals, healthy fats, and fiber. Filling diet with a variety of functional foods — including both conventional and fortified foods — can help to ensure availability of nutrients, and protect against nutrient deficiencies. In fact, since the introduction of fortified foods, the prevalence of nutrient deficiencies has significantly decreased around the globe. For instance, after iron-fortified wheat flour was introduced in Jordan, rates of iron deficiency anemia among children were nearly cut in half. Fortification has also been used to prevent other conditions caused by nutrient deficiencies, including rickets, goiter, and birth defects.

Protect against disease

Functional foods provide important nutrients that can help protect against disease.

Many are especially rich in antioxidants. These molecules help neutralize harmful compounds known as free radicals, helping prevent cell damage and certain chronic conditions, including heart disease, cancer, and diabetes.

Some functional foods are also high in omega-3 fatty acids, a healthy type of fat shown to reduce inflammation, boost brain function, and promote heart health.

Other types are rich in fiber, which can promote better blood sugar control and protect against conditions like diabetes, obesity, heart disease, and stroke. Fiber may also help prevent digestive disorders, including diverticulitis, stomach ulcers, hemorrhoids, and acid reflux.

Promote proper growth and development

Certain nutrients are essential to proper growth and development in infants and children.

For example, cereals, grains, and flours are often fortified with B vitamins like folic acid, which is essential for fetal health. Low levels of folic acid can increase the risk of neural tube defects, which can affect the brain, spinal cord, or spine. It's estimated that increasing the consumption of folic acid could decrease the prevalence of neural tube defects by 50–70%. Other nutrients commonly found in functional foods also play key roles in growth and development, including omega-3 fatty acids, iron, zinc, calcium, and vitamin B12.

Categories of Functional Foods

Functional foods can be categorized based on their origin into several types: conventional foods, modified foods, medical foods, and foods for special dietary use. Each type has distinct characteristics and purposes in dietary practices, aligning with specific health benefits or nutritional needs.

Conventional foods

These are naturally nutrient-rich foods that inherently contain beneficial compounds. Examples include fruits, vegetables, whole grains, nuts, and legumes. For instance, tomatoes and carrots are rich in phytochemicals like lycopene and beta-carotene, respectively, which are known for their antioxidant properties and potential to reduce the risk of certain chronic diseases.

Modified foods

These are standard foods enhanced through fortification or enrichment to boost their nutritional value. Common examples include orange juice fortified with calcium and vitamin D, bread enriched with folic acid, and cereals fortified with various vitamins and minerals. These modifications aim to address specific health concerns, such as improving bone health or preventing neural tube defects.

Medical foods

Designed specifically for dietary management of a disease or condition that has distinctive nutritional needs that cannot be met by normal diet alone. Medical foods are formulated to be consumed or administered under the supervision of a physician. An example includes formulas for patients with phenylketonuria (PKU), a genetic disorder requiring a diet low in phenylalanine, an amino acid that is harmful in high amounts to those with the condition.

Foods for special dietary use

Similar to medical foods, these are intended for specific dietary management of diseases or conditions but can be used without medical supervision. Examples include gluten-free foods for individuals with celiac disease, lactose-free dairy products for those with lactose intolerance, and low-sodium foods for people managing hypertension.

These foods help individuals avoid adverse reactions while meeting nutritional requirements.

Benefits of functional foods

Functional foods uniquely blend nutrition with enhanced health benefits, designed not only to nourish but also to offer preventive properties against chronic diseases. These foods are rich in essential nutrients, such as vitamins, minerals, and antioxidants, and provide health benefits that surpass those of conventional foods.

Nutritional benefits

Functional foods are primarily valued for their enhanced nutrient profiles. For example, cereals and juices fortified with vitamins and minerals, such as calcium and vitamin D, can significantly contribute to bone health. At the same time, foods enriched with antioxidants help combat oxidative stress, thus maintaining cellular health. These foods are engineered to include higher levels of specific nutrients that support overall well-being.

Health benefits

Beyond basic nutrition, functional foods can influence bodily functions positively. Probiotics found in yogurts and some cheeses, for instance, help maintain a healthy gut microbiota, which is

essential for effective digestion and immune function. Similarly, omega-3 fatty acids, prevalent in fatty fish and flaxseeds, are known for their anti-inflammatory effects, which are crucial for maintaining heart health and cognitive function. These components also aid in reducing symptoms of metabolic syndrome and other inflammation-related disorders.

Preventive benefits against chronic diseases

Perhaps the most significant value of functional foods lies in their potential to prevent lifestyle-related diseases. Foods high in dietary fibers, like whole grains and fortified bread, have been linked to lower risks of heart diseases due to their role in cholesterol management.

Similarly, the regular consumption of foods rich in phytochemicals—such as flavonoids in berries and green tea—has been associated with reduced risks of chronic diseases, including heart disease and diabetes. These effects are attributed to their ability to modulate metabolic pathways and reduce inflammation. The relationship between diet and disease prevention is further supported by numerous studies, underscoring the role of functional foods in a health-conscious consumer's diet. Integration of these foods into daily dietary practices is advocated not only for health enhancement but also as a strategy for disease risk reduction, particularly for chronic conditions prevalent in the aging population.

Uses of Functional Foods

Functional foods are not just beneficial for their nutrient content; they also play a crucial role in promoting health and preventing diseases when incorporated into daily diets and specific therapeutic regimens.

Understanding how to integrate these foods into daily meals can maximize their health benefits.

Daily dietary integration of functional foods

Integrating functional foods into daily meals is pivotal for maintaining long-term health. Simple incorporations, like starting the day with a bowl of fortified cereal or adding chia seeds rich in omega-3 fatty acids to your morning smoothie, can enhance dietary nutrient profiles.

Snacking on nuts or berries, or using olive oil for cooking, are other easy ways to include functional ingredients that support heart health and provide anti-inflammatory benefits.

Specific uses in medical or dietary therapies

Functional foods often play a specific role in dietary therapies for managing health conditions. For instance, the inclusion of oats and barley, which are high in soluble fiber, can aid in managing cholesterol levels under medical guidance. Foods fortified with plant sterols are recommended for patients with heart disease risks, as they help lower LDL cholesterol levels.

Probiotics and fermented foods are prescribed to improve gut health and aid digestion, particularly in patients suffering from gastrointestinal disorders.

How to incorporate functional foods into various meals

1. Breakfast

Opt for high-fiber cereals or breads enriched with vitamins and minerals. Adding fruits like blueberries or strawberries can provide antioxidants for a nutritious start to the day.

2. Lunch and dinner

Include at least one portion of vegetables or salad with dressings that include flaxseed oil or virgin olive oil. Incorporate lean proteins such as fish or poultry that are cooked in healthy fats and seasoned with turmeric or garlic, which are known for their anti-inflammatory properties.

3. Snacks

Choose nuts, seeds, or yogurts with live cultures, which are excellent for snacking while providing functional benefits such as improving heart health and digestive health.

4. Beverages

Replace sugary drinks with green tea or functional juices like cranberry juice, which supports urinary tract health.

Examples of functional foods

Functional foods are nutritional powerhouses that go beyond mere sustenance, providing health benefits and playing a significant role in disease prevention.

Here are detailed descriptions of some functional foods and their unique advantages:

1. Omega-3 enriched eggs

These eggs are fortified with omega-3 fatty acids, typically sourced from fish oil or flaxseed. Omega-3s are crucial for heart health, reducing inflammation, and supporting brain function. Regular consumption of omega-3 enriched eggs can contribute to reduced blood pressure and lower triglyceride levels.

2. Probiotic yogurts

Containing live beneficial bacteria like Lactobacillus and Bifidobacterium, these yogurts help in maintaining gut health, enhancing immune response, and potentially reducing the risk of chronic diseases such as irritable bowel syndrome and infectious diarrhea.

3. Fortified cereals

Typically enriched with vitamins and minerals such as iron, calcium, and B vitamins, fortified cereals can help address nutrient deficiencies and contribute to improved overall health. These cereals play a crucial role in metabolic processes and can aid in preventing conditions like anemia.

4. Chia seeds

Although less mainstream, chia seeds are a robust source of fiber, omega-3 fatty acids, and minerals.

Their consumption is linked to improved cardiovascular health and digestion. They can be easily incorporated into diets through smoothies, yogurts, or as a topping on salads.

5. Kefir

A fermented milk product similar to yogurt but containing different types of beneficial bacteria. Kefir supports digestive health and may improve lactose digestion. Its probiotic qualities also contribute to a healthy immune system.

6. Soy nuts

Made from whole soybeans that have been soaked and then baked or roasted, soy nuts are a crunchy, nutritious snack. They are rich in protein, fiber, and isoflavones, which may help in reducing the risk of heart disease and improving bone health.

Each of these functional foods offers unique benefits that contribute to specific health outcomes. Whether through direct nutrient provision, disease risk reduction, or supporting overall wellness, the incorporation of functional foods into daily diets can play a pivotal role in maintaining and enhancing health.

Risks and side effects of functional foods

While functional foods offer numerous health benefits, from enhancing nutrient intake to reducing disease risks, they also come with potential concerns that consumers and healthcare providers should consider. The most significant risks associated with functional foods relate to the safety and unintended effects of bioactive components when consumed in large quantities or without proper guidance.

Adverse effects and interactions

Functional foods enriched with nutrients or bioactive compounds can interact with medications or other dietary components in ways that may not be fully predictable. For example, excessive intake of fortified products could lead to nutrient imbalances or toxicities. High doses of fat-soluble vitamins, like vitamin A or D, can lead to hypervitaminosis, which manifests in symptoms ranging from liver damage to skeletal abnormalities.

Allergic reactions and food sensitivities

Some functional foods may trigger allergic reactions or sensitivities due to their enhanced ingredients or the introduction of novel proteins to which consumers may not be accustomed. For instance, people with allergies to dairy or soy could inadvertently consume these allergens present in functional foods not typically associated with these ingredients.

Long-term health impacts

The long-term health impacts of consuming high concentrations of certain bioactives are still not well understood. While these components are beneficial in moderation, their concentrated forms in functional foods could have unforeseen health consequences over prolonged periods. For example, phytoestrogens from soy are beneficial in moderate amounts but could potentially promote the growth of hormone-sensitive cancers if consumed excessively.

Regulatory and labeling issues

There is also the issue of regulation and transparency in the labeling of functional foods. Without stringent regulatory oversight, manufacturers may not always provide clear or accurate information regarding the concentrations of bioactive components and their potential health effects. This can lead to confusion and misuse by consumers who might not be aware of the appropriate quantities or conditions under which these foods should be consumed.

Consumer education and misuse

Finally, there is a risk associated with the consumer's ability to integrate functional foods appropriately into their diet. Misunderstandings about the health benefits of functional foods can lead to over-reliance on these products at the expense of a more balanced dietary approach.

Ensuring that consumers are well-educated about the realistic benefits and potential risks of functional foods is crucial for their safe and effective use.

Nutritional Assessment

The British Dietetic Association (BDA) defines nutritional assessment as the systematic process of collecting and interpreting information in order to make decisions about the nature and cause of nutrition-related health issues that affect an individual. It can be done by a healthcare professional or self-assessment using online tools.

Nutritional assessment is the interpretation of anthropometric, biochemical (laboratory), clinical and dietary data to determine whether a person or groups of people are well nourished or malnourished (overnourished or undernourished). Nutritional assessment can be done using the ABCD methods. These refer to the following:

- A. Anthropometry
- B. Biochemical methods
- C. Clinical methods
- D. Dietary methods

Importance of Nutritional Assessment

You are what you eat. Committing to nutritional assessment helps you know what you should and should not be eating if you want to live a healthy life. Let's look at some other reasons why you should prioritize nutritional assessment.

1. Nutritional assessment helps people understand their own dietary intake and how it compares with the recommended daily allowances for nutrients.
2. Regular nutritional assessment allows you to identify any potential risks associated with poor nutrition.
3. It helps people make informed decisions about changes to their diets.
4. A nutritional assessment provides information about whether or not there are specific foods that you shouldn't eat.
5. It helps you learn how to plan meals and snacks ahead so you don't have to rely on fast food or convenience options.
6. Regular nutritional assessment is the only way to ensure you're getting enough nutrients from your meals and in the right quantities.

Objectives of Nutritional Assessment

The objectives of a nutritional assessment depend on the context of the program and what you want to achieve. In the case of a one-on-one program with an individual, the common goal should be improving the health habits and overall lifestyle of the patient. Nutritional assessment should also identify and address any cases of possible malnutrition. Other common objectives are:

1. Nutritional assessment evaluates a person's overall health and nutritional status.
2. It identifies possible nutrient deficiencies in an individual
3. Nutritional assessment allows the experts to evaluate the effectiveness of prescribed treatments.
4. It's an effective way to monitor progress toward goals set during treatment.

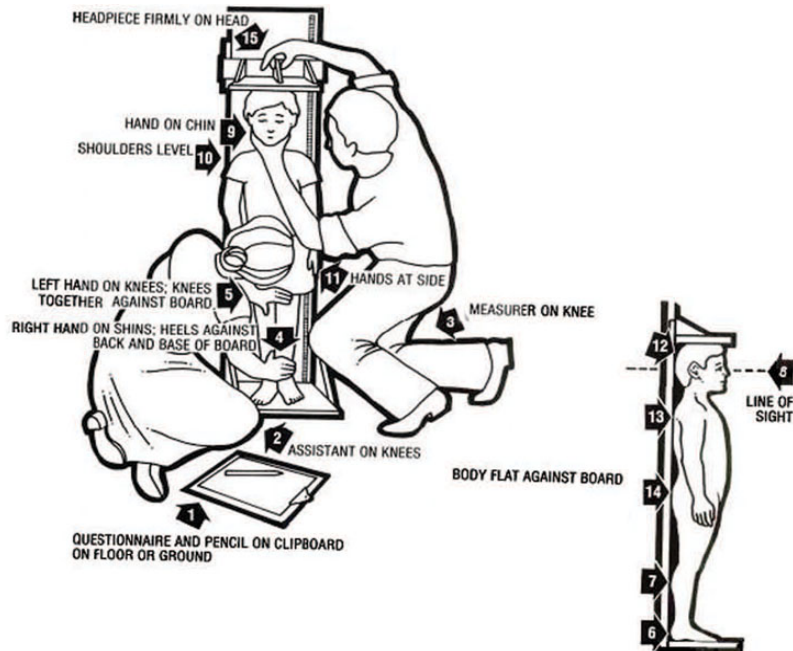
5. It helps you to prevent malnutrition.
6. It provides an opportunity for the experts to educate their patients about proper nutrition.
7. Nutritional assessment promotes healthy lifestyles.
8. It encourages compliance with recommendations for treatment.

1. Anthropometric Nutritional Assessment

Anthropometric measurements are non-invasive quantitative measurements of the body that provide valuable assessments of the nutritional status of children and adults. Typically, it involves the measurement of the size, weight, and proportions of the body.

Common anthropometric measurements include:

1. Waist Circumference
2. Height
3. Weight
4. Skinfold thickness



**Advantages**

1. It uses simple, safe, and non-invasive procedures.
2. Anthropometric assessment techniques can be applied to a large sample size
3. It is objective with high sensitivity and specificity.
4. It can be done by healthcare providers without specialized training.

Disadvantages

1. An anthropometric assessment covers limited nutritional diagnosis.
2. Anthropometric measurements cannot identify protein and micronutrient deficiencies or detect small disturbances in nutritional status.

Weight for age

The development of a child is determined by the increase in weight over a given time. This in turn is determined by weight measurements over regular intervals. Here the child weight is compared with reference weight value for his age. The most widely used method is that of Gomez and Co-workers (1956). The observed weight of a child is expressed as a % of expected weight of a child of that age using the 50th percentile (medium) of Havard Standards.

Gomez Classification

% expected weight for age	Classification	Category of nutritional status	
>90%	Normal	Normal	
76-90%	Mild malnutrition	1 st	degree
malnutrition			
61-75%	Moderate malnutrition	2 nd	degree
malnutrition			
< 60%	Severe Malnutrition	3 rd	degree
malnutrition			

In 1972 nutrition sub-committee of Indian Academy of Paediatrics (IAP) proposed another classification.

% of expected water	classification
>80%	Normal
71 – 80%	Grade I
61 – 70%	Grade II
51 – 50%	Grade III severe malnutrition
< 50%	Grade I severe malnutrition

Disadvantage for the 2 classifications

1. It takes no account of other measurements
2. It assumes that children are of the same height
3. Single measurement of weight does not indicate definite malnutrition when former weight measurement are not known
4. Harvard references are not easily available

Advantages:

- (1) It is very simple to measure.
- (2) It is widely used.

Height for age:

It is used as an indication of nutritional status of groups of population for estimating past and chronic malnutrition but not necessarily the present nutritional status. The disadvantage is that the deficiency in height takes some time to occur and it may not be manifested in malnourished infants. Also, genetical differences are partly responsible for the variation in height in any group in any population.

Weight for height

This can be expressed as a % of the reference median weight for median height at any age.

Advantage

The advantage for using weight for height as an index of nutritional status is its apparent age independence, this allows its use in populations where ages are uncertain or unknown. Marked decrease in weight for height is a more reliable finding in the determination of PEM in all age groups.

Disadvantages

The difficulty in measuring body length in young infants which may make it difficult to obtain adequate data in this age group.

Mid Arm Circumference (MAC)

One of the most widely used indices for the assessment of nutritional status especially during childhood because the tape used is inexpensive and portable.

Advantage

Measurement is easy and simple, takes less time to perform. It involves only a simple measurement, can be taught to lay people, equipment is inexpensive. MAC correlates well with weight and weight for height. It indicates state of muscle protein. MAC should be measured only when weighing scale, cannot be supplied.

Cut off points:	AC > 14cm	normal nutritional state
	AC 14 – 12.5	mild/modest undernutrition
	AC < 12.5	severe undernutrition

Body Mass Index

In adult low weight for height may indicate inadequate nutrition whereas high weight for height may indicate overweight and obesity. In recent years BMI is used. It is the best simple and quantitative anthropometric indicator of body composition and thus nutritional status.

BMI Formula

To calculate your ideal weight, you can use the following BMI formula:

$$\text{BMI} = \text{Weight (kg)} / (\text{Height (m)} \times \text{Height (m)})$$

You can also calculate your BMI using BMI calculators available on various websites or digital applications.

Example of Calculating BMI

Here is an example of calculating BMI with a weight of 65 kg and a height of 170 cm or 1.70 m:

- Multiply your height in meters = $1.70 \text{ m} \times 1.70 \text{ m} = 2.89 \text{ m}^2$.
- Then, divide your weight by the square of your height = $65 \text{ kg} : 2.89 \text{ m}^2 = 22.49 \text{ kg/m}^2$.

Thus, your BMI is 22.49 kg/m^2 .

Based on the BMI classification from both WHO and the Indonesian Ministry of Health, this value indicates that your weight is within the normal or ideal range.



Classification	BMI(kg/m ²)	
	Principal cut-off points	Additional cut-off points
Underweight	<18.50	<18.50
Severe thinness	<16.00	<16.00
Moderate thinness	16.00 - 16.99	16.00 - 16.99
Mild thinness	17.00 - 18.49	17.00 - 18.49
Normal range	18.50 - 24.99	18.50 - 22.99
		23.00 - 24.99
Overweight	≥25.00	≥25.00
Pre-obese	25.00 - 29.99	25.00 - 27.49
		27.50 - 29.99
Obese	≥30.00	≥30.00
Obese class I	30.00 - 34.99	30.00 - 32.49
		32.50 - 34.99
Obese class II	35.00 - 39.99	35.00 - 37.49
		37.50 - 39.99
Obese class III	≥40.00	≥40.00

Source: Adapted from WHO, 1995, WHO, 2000 and WHO 2004.

SKINFOLD THICKNESS

The most direct measure of fatness in people is measurement of skinfold thickness using skinfold calliper. These springs loaded callipers exert a constant pressure on a fold of skin, the thickness of skin is indicated in a meter. The thickness depends in the amount of fat stored subcutaneously in the region of the skinfold. It is measured at several sites and it is still the representative of the total amount of body fat. Typically, it is determine at 4 sites

- (i) over the triceps muscle
- (ii) over the biceps
- (iii) in the subscapular region
- (iv) in the supra-iliac region.

The single triceps muscle is sometimes used in nutritional survey because it can be measured quickly.

2. Biochemical Assessment

Biochemical assessment involves checking the level of nutrients in a person's blood, urine, or stool, usually through a lab test. These lab tests can help a trained nutritionist/medical practitioner discover any medical problems affecting patient nutritional status or appetite. For example, a lab scientist might take your blood sample to measure the level of glucose in your body.

During a full biochemical assessment, the physician will screen the following biochemical parameters: albumin, prealbumin, transferrin, hemoglobin, urea and creatine, lymphocytes, and point deficiencies.

**Advantages**

1. It shows earliest indication of malnutrition or any nutritional deficiencies in the body.
2. Biochemical assessments also confirm the clinical diagnosis of nutritional status and/ or risk for a disease.

Disadvantages

1. It is time-consuming.
2. The health practitioner needs to run multiple biological tests for a proper diagnosis.

3. Clinical Nutritional Assessment

In addition to the anthropometric assessments, you can also assess clinical signs and symptoms that might indicate potential specific nutrient deficiency. Special attention are given to organs such as skin, eyes, tongue, ears, mouth, hair, nails, and gums. Clinical methods of assessing nutritional status involve checking signs of deficiency at specific places on the body or asking the patient whether they have any symptoms that might suggest nutrient deficiency.

Advantages

- It helps the health practitioner dictate changes in the body's metabolism.

Disadvantages

- It is expensive.
- It only provides limited data on food composition.

4. Dietary Assessment

Dietary assessment is the process of collecting information about what a person eats and drinks over a period of time. In other words, it is a record of the foods one eats in an attempt to calculate their potential nutrient intake.

During a dietary assessment, the health practitioner analyzes the energy, nutrients, and other dietary constituents using food composition tables.

The goal of dietary assessment is to identify appropriate and actionable areas of change in the patient's diet and lifestyle and to improve the overall wellbeing of the patient. For a detailed analysis, the health practitioner can deploy one or more of these methods:

- Diet Record
- 24-hour recall
- Food Frequency Questionnaire

Advantages of Dietary Assessment

- It provides contextual information about a person's nutritional intake.
- Results from the dietary assessment are largely accurate due to more detailed descriptions of foods and portion sizes.

Disadvantages of Dietary Assessment

- It relies on accurate recall of dietary intake over a long period of time.
- It is prone to misreporting, especially when the health practitioner adopts food frequency questionnaires for data gathering.

This usually involves two Methods, i.e., a group method, or dietary intake of an individual. It is determined by record or recall of all foods consumed over a specified period of time. This is the most commonly used method for the field surveys.

Group Method

1 Food Balance sheet

On the basis of food availability food balance sheet from the entire population is prepared. The mean annual amount per person is obtained by dividing the total amount of different foods with the total population of the area. The mean intake of different nutrients is an essential part of food balance sheets.

Advantages

1. The food balance sheet gives view of the total food supplies of a community. It indicates whether food consumed by the population is inadequate, adequate or more than adequate.
2. It seems as a basis for planning of food programme and for emergency rationing of food
3. Valuable in inferring general food habits of the people

Disadvantages

1. The reliability of such data depends on the statistics used for calculation and level of development of the country.
2. The data shows the total amount of food available and not the actual amounts consumed
3. Longitudinal differences in food consumption are not reflected for mean total consumption data of the whole population.

Food Accounts

This method of dietary assessment is commonly used for subjects living in institutions, families or groups. It involves detailed recording of the amount of food consumed over a period of time. This

is compiled by an inventory of food supplies both at the start and end of the survey. The method provides accurate information on food consumption.

Advantages

1. Larger samples can be obtained and food consumption data for longer period of time can be collected.
2. As seasonal variation are taken into account, the method provides excellent information on the annual mean food consumption
3. The method is cheap and does not alter the diet of the subjects to a great extent

Disadvantage

1. Families or household are not always representative of the whole population
2. It can be used only with the likely population
3. Precision may decrease after four days
4. Food distribution within the family is not known
5. The records may not always reveal how much food was actually consumed or thrown away due to spoilage or plate wastage

Individual Dietary Intakes

Dietary data on individuals is collected to obtain more precise measurement of average nutrient intake and to determine inadequacies if any. Assessment ranges from a qualitative type of inquiry to those of a more quantitative nature. It is one of the most accurate methods and is referred to as Precise and Weighed Individual inventory method. An inventory of the food supply both at the beginning and end of survey is made. As more food is acquired it is weighed and recorded. Weights and recording of food consumption at home and outside and food wastage are maintained. At the end, the amount of food wasted is added to the amount of food left over to obtain total wastage. Then divided the total food consumed by the number of adults given daily food with consumption per person. i.e. (initial inventory + issues or purchases) – (final inventory + waste)

$$\text{Food consumed/person/day} = \frac{\text{Total weight of food item consumed}}{(\text{days of survey} \times \text{number of persons fed daily})}$$

Advantages

1. The information can be collected by the subjects themselves and require minimal supervision
2. The amount consumed can be recorded accurately

Disadvantages

1. The sample size is not representative as whinke in are selected and the results cannot be generalised.
2. It is costly and requires trained personal
3. It changes the diet of the respondents so that it does not represent normal consumption pattern.

Interview method

This is to collect food consumption data

- (1) Diet recall

(2) Diet history

Diet Recall

Food consumption for a specified period of time prior to the survey can be recalled, and it is often referred to as the 24 hours recall method.

Individual intake (in volume) X Raw amount = Y

Total cooked quantity (in volume)

Standardised volumetric measures of the ingredient are recorded

Advantage

Useful method is recapitulation of one's habitual diet It is helpful in revealing extreme daily variation in the diet

Disadvantage

1. A day intake may not represent usual intake.
2. Estimate becomes difficult when diet has a lot of variety.
3. Subjects reporting may not be sure about intake.

Diet History

Provide a more comprehensive assessment of diet and it permits investigation of lesser known or unidentified dietary factor that can be retained for future examination. The normal daily dietaries are first recorded along with the left-over of each meal, the composition, snacking etc. Seasonal variations are also included.

Advantage

It is inexpensive and convenient

The representative and large sample size / events

Disadvantages:

It demands greater comportsment, personal character of the investigator.

Does not give precise data on individual food consumption.

Diet histories are subject to problems of recall.

Nutrient uptakes tend to be overestimated especially from trace elements.

Food frequency method

Intakes in terms of frequency with which various foods items are consumed is recorded.

Questionnaire method

It is identical to diet history. Questionnaires are sent to the respondents to fill and return.

Advantage:

1. It is possible to collect data in large samples in short period with a small budget.
2. Random samples can be used.

Food composite analysis for laboratory estimate

This method involves sampling of each item during meals with subsequent blending of representative samples and analysis for various nutrients.

Food fortification

Food fortification refers to the addition of micronutrients to processed foods. In many situations, this strategy can lead to relatively rapid improvements in the micronutrient status of a population, and at a very reasonable cost, especially if advantage can be taken of existing technology and local distribution networks. Since the benefits are potentially large, food fortification can be a very cost-effective public health intervention. However, an obvious requirement is that the fortified food(s) needs to be consumed in adequate amounts by a large proportion of the target individuals in a population. It is also necessary to have access to, and to use, fortificants that are well absorbed yet do not affect the sensory properties of foods. In most cases, it is preferable to use food vehicles that are centrally processed, and to have the support of the food industry. Fortification of food with micronutrients is a valid technology for reducing micronutrient malnutrition as part of a food-based approach when and where existing food supplies and limited access fail to provide adequate levels of the respective nutrients in the diet. In such cases, food fortification reinforces and supports ongoing nutrition improvement programmes and should be regarded as part of a broader, integrated approach to prevent MNM, thereby complementing other approaches to improve micronutrient status.

Food Fortification and Enrichment

Food fortification is the addition of micronutrients (essential trace elements and vitamins) to food products. Food enrichment specifically means adding back nutrients lost during food processing, while fortification includes adding nutrients not naturally present.

Food manufacturers and governments have used these practices since the 1920s to help prevent nutrient deficiencies in populations. Common nutrient deficiencies in a region often result from local soil conditions or limitations of staple foods. The addition of micronutrients to staples and condiments can prevent large-scale deficiency diseases in these cases.

Food fortification has been identified as the second strategy of four by the WHO and FAO to begin decreasing the incidence of nutrient deficiencies at the global level. As outlined by the FAO, the most commonly fortified foods are cereals and cereal-based products; milk and dairy products; fats and oils; accessory food items; tea and other beverages; and infant formulas.

Types

Fortification is present in common food items in two different ways: adding back and addition. Flour loses nutritional value due to the way grains are processed; enriched flour has iron, folic acid, niacin, riboflavin, and thiamine added back to it. Conversely, other fortified foods have micronutrients added to them that don't naturally occur in those substances. An example of this is orange juice, which often is sold with added calcium.

Food fortification can also be categorized according to the stage of addition:

1. Commercial and industrial fortification (wheat flour, corn meal, cooking oils)
2. Biofortification (breeding crops to increase their nutritional value, which can include both conventional selective breeding, and genetic engineering)

3. Home fortification (example: vitamin D drops)

Supplementation

Supplementation is the term used to describe the provision of relatively large doses of micronutrients, usually in the form of pills, capsules or syrups. It has the advantage of being capable of supplying an optimal amount of a specific nutrient or nutrients, in a highly absorbable form, and is often the fastest way to control deficiency in individuals or population groups that have been identified as being deficient.

In developing countries, supplementation programmes have been widely used to provide iron and folic acid to pregnant women, and vitamin A to infants, children under 5 years of age and postpartum women. Because a single high-dose vitamin A supplement improves vitamin A stores for about 4–6 months, supplementation two or three times a year is usually adequate. However, in the case of the more water-soluble vitamins and minerals, supplements need to be consumed more frequently. Supplementation usually requires the procurement and purchase of micronutrients in a relatively expensive pre-packaged form, an effective distribution system and a high degree of consumer compliance (especially if supplements need to be consumed on a long-term basis). A lack of supplies and poor compliance are consistently reported by many supplementation programme managers as being the main barriers to success.